

Shafei Waqar

Maple Ridge, BC | 604-446-2238 | shafei.waqar@gmail.com | linkedin.com/in/ShafeiWaqar | github.com/ShafeiW

TECHNICAL SKILLS

Languages: Python, C/C++ , SQL, JavaScript, Typescript, HTML/CSS

Technologies: Claude Code, Codex, AWS (S3, Step Functions, Cloudwatch), Bootstrap, React, React Native, Node.js, MongoDB, Flask, PyTorch, OpenAI API, Selenium, BeautifulSoup, Cloudinary, Git, GitHub, Docker, FastAPI, Streamlit, uv

Relevant Coursework: Data Structures and Algorithms, Intro to Software Engineering, Introduction to AI, Computer Systems, Systems Programming

EXPERIENCE

Software Engineering Intern

Aug. 2025 – Apr. 2026

Marr Labs (YC W24)

Remote/San Francisco, CA

- Engineered voice agent evaluation pipeline and latency tooling for agents deemed for Rocket Mortgage and additional customers, supporting **500K–1M monthly calls**, cutting startup times by **90%** and extending with PHI-compliant coverage for healthcare customers
- Automated a legacy **TypeScript** system built on internal API for generating evaluation agents using **OpenAI API + GPT-5** and **o3** models, eliminating manual configuration and significantly increasing evaluation startup efficiency
- Built and scaled an LLM-powered pipeline to analyze and classify live call transcripts, improving classification accuracy by **10%** across growing customer workloads
- Delivered **30+** prompt optimizations for agent responses and evaluation criteria, reducing false positive classifications by **25%** and improving response accuracy by **15%**

Software Developer

Nov. 2024 – Present

PreMedTalk Blog

Remote

- Architected and launched a full-stack **Next.js** blogging platform with rich-text editing and dynamic HTML content storage in **MongoDB**, streamlining content management for pre-medical students
- Revitalized the user interface and built an intuitive admin dashboard, with easy content upload with **Cloudinary**, boosting usability and driving higher engagement
- Partnered with club leaders to align platform features with organizational objectives, resulting in increased user satisfaction and platform adoption

PROJECTS

Low-Light Image Enhancer | Python, PyTorch, FastAPI, Streamlit, uv

[Github](#)

- Reproduced **Zero-DCE** (Guo et al., CVPR 2020) in PyTorch and improved best training loss by **9.6%** ($0.0929 \rightarrow 0.0840$) over paper defaults by tuning loss weights across 3 training runs on 485 images
- Extended the paper's fixed **8-iteration** design with a learned IterationPredictor branch (adaptive range **4–12**) using differentiable soft gating, with a luminance safety clamp that reduced predicted iterations from **12.00 to 4.19** on bright images to prevent over-exposure
- Served the **79K-parameter** model via a FastAPI endpoint with adjustable enhancement strength (**0.1–3.0×**) and a Streamlit frontend with live per-image iteration count display

Amazon Review Analyzer | Python, Flask, PyTorch, Selenium, BeautifulSoup, HTML5, CSS3

[Github](#)

- Developed Amazon Review Analyzer, an end-to-end sentiment analysis system built with **PyTorch** and **Flask**, achieving **92.4%** validation accuracy
- Fine-tuned a **BERT** model on Amazon reviews dataset, implementing custom preprocessing and achieving **0.939** F1 macro score on a 5-class sentiment classification task, demonstrating high model reliability and performance
- Built a resilient web-scraping framework with **Selenium** and **BeautifulSoup**, successfully navigating Amazon's anti-bot defenses to harvest product data
- Built a user-friendly, responsive web interface using modern **JavaScript**, **HTML5**, and **CSS3**, facilitating seamless interaction with the sentiment analysis backend

EDUCATION

Simon Fraser University

Burnaby, BC

Bachelor of Science in Computing Science, Minor in Mathematics

Sep. 2023 – May 2027